

# Building Your First Accessible Gutenberg Block

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WordPress Accessibility Meetup

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# Today's Goal

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Take the mystery out of the WordPress Block Editor.

Shortcodes and  
"Classic" development  
feel outdated.



Build a real custom  
Gutenberg block  
from scratch —  
step by step.

# What We'll Cover

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- Setting up a local development environment
- Using `@wordpress/create-block` to scaffold a project instantly
- Understanding the basic file structure: `block.json`, `edit.js`, `save.js`
- Adding attributes so users can change text within the editor
- The Edit interface vs the Save function
- JSX basics in the WordPress context

# By the End of This Session, You Will:

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- Understand the difference between Edit (admin view) and Save (visitor view)
- Know how to use JSX basics within WordPress
- Generate, compile, and activate your own custom block plugin
- Feel confident moving from Classic to Block-based development

# What Is a Gutenberg Block?

A self-contained, reusable piece of content in the WordPress block editor.

## Old way — Shortcodes

- [contact-form]
- [gallery ids='1,2,3']
- [button text='Click me']
- Memorise syntax. No preview. Breaks if plugin deactivates.

## New way — Blocks

- Visual drag-and-drop in the editor
- Live preview while editing
- Accessible HTML output
- You own 100% of the code

# The Two Files That Matter Most

## `edit.js`

What the **ADMIN** sees  
in the **WordPress** dashboard

- Rich controls and toolbars
- Colour pickers and toggles
- Live preview of content

vs

## `save.js`

What **VISITORS** see  
on the live website

- Clean semantic HTML
- Static and deterministic
- Saved to the database

# Setting Up Your Environment

You need two free things installed before we begin:

## 1. Node.js

[nodejs.org/en/download](https://nodejs.org/en/download)  
Choose the LTS version.

## 2. LocalWP

[localwp.com](https://localwp.com) (free)  
Runs WordPress on your machine.

**Verify Node.js is installed:**

```
$ node --version    # shows v18.x.x or higher
$ npm --version     # shows v9.x.x or higher
```

# One Command Creates Everything

Navigate to your WordPress plugins folder, then run:

```
$ cd wp-content/plugins
$ npx @wordpress/create-block my-notice-block
$ cd my-notice-block
$ npm start

# npm start watches your files and recompiles automatically
```

## What is npx?

npx comes with Node.js. It downloads and runs a tool without installing it first — so `npx @wordpress/create-block` downloads the official WP block scaffolding tool and runs it immediately.



# Understanding the File Structure

`block.json`

Block ID card — name, title, attributes, supports. WordPress reads this first.

`src/edit.js`

Editor view — what the admin sees. Import WP tools, export Edit component.

`src/save.js`

Frontend view — static HTML saved to the database. Use `useBlockProps.save()`.

`src/style.scss`

Shared styles — applied in both the editor and on the live website.

## Adding Attributes — Making It Editable

## Step 1 — block.json

```
"noticeText": { "type":
"string", "default": "Your
notice here..." },
  "noticeType": { "type":
"string", "default": "info"
}
```

**Key concept: attributes flow one way**

WordPress passes values in. You call `setAttributes()` to update them. The block re-renders automatically.

## Step 2 — edit.js

```

Edit (
  { attributes,
  setAttributes }
) {
  return (
    <RichText tagName='p'
value={attributes.noticeText
}
    onChange={ (v) =>
setAttributes ({noticeText:v
}) }
    />
  ) ;
}

```

# The Golden Rule: Edit vs Save

`edit.js`

Admin view  
inside the  
WordPress dashboard

- Rich controls and pickers
- Drag handles
- Live preview

VS

`save.js`

Visitor view  
on the live  
website

- Clean HTML output
- Saved to database
- Accessible markup

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# JSX Basics in the WordPress Context

JSX looks like HTML but it's JavaScript. WordPress compiles it with npm start.

```
// JSX in edit.js - looks like HTML, it's JavaScript
return (
  <section
    role="region"
    aria-label={`${noticeType} notice`}
  >
    <span aria-hidden="true">{ICONS[noticeType]}</span>
    <RichText tagName='p'
      value={noticeText}
      onChange={ (v) => setAttributes({ noticeText: v })}
    />
  </section>
);
```

# Accessibility Built Into Your Block

`role="region"`

Turns the wrapper into a navigable landmark — screen reader users can jump to it.

`aria-label="..."`

Gives the landmark an accessible name — tells users exactly what region they entered.

`aria-live="polite"`

Announces dynamic changes without interrupting screen reader users.

`aria-hidden="true"`

Hides decorative icons — prevents duplicate announcements from screen readers.

`<section>` not `<div>`

Semantic HTML — a `<div>` means nothing to AT; `<section role=region>` is a navigable landmark.

# Activate and Test Your Block

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1 Go to wp-admin → Plugins → find 'My Notice Block' → Activate

2 Create a new Page or Post in WordPress

3 Type / in the editor → search 'notice' → click to insert

4 Select the block → open the Settings sidebar (⚙ icon)

5 Change the Notice Type dropdown → watch the icon update live

# Your Block vs Major Plugins

| Feature                | Your Block | Yoast SEO | WooCommerce | Elementor | ACF      |
|------------------------|------------|-----------|-------------|-----------|----------|
| Custom HTML output     | ✓          | Partial   | Partial     | ✗         | Partial  |
| Zero plugin dependency | ✓          | ✗         | ✗           | ✗         | ✗        |
| WCAG 2.1 AA built in   | ✓          | Partial   | Partial     | Partial   | Partial  |
| You own the code       | ✓          | ✗         | ✗           | ✗         | ✗        |
| No monthly fee         | ✓          | Freemium  | Freemium    | Freemium  | Freemium |
| Semantic HTML output   | ✓          | Mostly    | Mostly      | ✗         | Mostly   |

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# From Classic to Block-Based Development

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## You Don't Need to Rewrite Everything Overnight

Start with:

- One new block for your next client project
- The create-block tool handles all the boilerplate
- Accessibility is already baked in — don't bolt it on later
- The same skills transfer to Full Site Editing and Block Themes



# Your Block Accessibility Checklist

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- ✓ Wrapper uses `<section role="region" aria-label="...">` not `<div>`
- ✓ `aria-live="polite"` if content updates dynamically
- ✓ `aria-hidden="true"` on all decorative icons
- ✓ All text contrast meets 4.5:1 (verified with WebAIM checker)
- ✓ All interactions work with Tab, Enter, and Space keys
- ✓ Tested with NVDA + Firefox or VoiceOver + Safari
- ✓ Never uses `*:focus { outline: none }`
- ✓ Inform users of notice type via colour + icon + aria-label (not colour alone)

# Where to Start

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## Recommended First Steps



Install Node.js ([nodejs.org](https://nodejs.org)) and LocalWP ([localwp.com](https://localwp.com))



Run `npx @wordpress/create-block` and explore the generated files



Edit `edit.js` and `save.js` following the patterns in this session



Test keyboard navigation: Tab through your block without a mouse



Run axe DevTools ([deque.com/axe](https://deque.com/axe)) to scan for accessibility issues

# Resources

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- **Block Editor Handbook** [developer.wordpress.org/block-editor/](https://developer.wordpress.org/block-editor/)
- **@wordpress/create-block** [developer.wordpress.org/block-editor/reference-guides/packages/packages-create-block/](https://developer.wordpress.org/block-editor/reference-guides/packages/packages-create-block/)
- **WordPress Accessibility Meetup** [equalizedigital.com/wordpress-accessibility-meetup/](https://equalizedigital.com/wordpress-accessibility-meetup/)
- **WCAG 2.1 Quick Reference** [www.w3.org/WAI/WCAG21/quickref/](https://www.w3.org/WAI/WCAG21/quickref/)
- **axe DevTools** [deque.com/axe/devtools/](https://deque.com/axe/devtools/)
- **WebAIM Contrast Checker** [webaim.org/resources/contrastchecker/](https://webaim.org/resources/contrastchecker/)
- **LocalWP (free local WordPress)** [localwp.com](https://localwp.com)

# Q&A

**Thank you!**

## Questions?

**Emilio Dominguez**

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